

SmartCare v0.1 - Initial Engineering Brief and AI Activity Card

A2 Case Study Stage 1 student resource

SmartCare scenario

SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The client says: 'We need software to help manage patients, practitioners and appointments.' This is not yet a complete specification.

Initial Engineering Brief

1. Problem summary

SmartCare Community Clinic currently uses a combination of spreadsheets and paper records to manage patients and appointments. These manual processes make it difficult to maintain accurate and consistent information and can create problems when managing appointments and locating patient records. The clinic has identified a need for software to help manage patients, practitioners and appointments more effectively. However, the client's current statement does not provide enough detail to define the complete system requirements. Further information is needed from SmartCare about the users, required information, appointment processes, security requirements and specific problems the new software must address before development begins.

2. Initial stakeholders

Stakeholder	Possible need
Patients	Easy access to booking information
Doctors	Access to patients records and appointment schedule
Reception Staff	Efficient patient search, appointment booking and status management
Clinic Owner	System that is cheap, efficient and easy to use while protecting patient information.

3. Initial features

Feature	Confirmed or provisional?	Why?
Practitioner Schedule View	Confirmed	The client said they wanted software to help manage practitioners and time.
Appointment Management	Confirmed	The clinic said they wanted software to help manage appointments
Patient Search	Confirmed	Clinic has stated that has trouble locating client files, this would help them locate them and keep it organised
Online appointment booking	Provisional	This could improve appointment

		management, but the client hasn't stated if they want it implemented
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4. Questions for the client

1. What information will be stored for each client?
2. Who is going to use the system and how often?
3. Should patients be able to cancel and book their own appointments?
4. Should the practitioners be able to book and cancel appointments themselves?
5. How far in advance should appointments be able to be booked?

5. What we do not yet know

1. We do not yet know what patient, practitioner and appointment information needs to be stored and for how long.
2. We do not yet know the exact requirements and functions each user needs from the system.
3. How much demand the clinic is expecting, so how many system would need to be in use?

AI Activity Card - Ask, Check, Explain

Before AI

What do I think the code does? What problems can I already identify?

AI request

Act as a tutor. Explain this code and identify potential problems. Do not provide a complete replacement. Ask me questions that help me reason about the solution.

Evaluate

Suggestion	Useful	Unclear	Incorrect	Out of scope

Decide

For each significant suggestion: Accept / Modify / Reject / Keep unverified.

Verify

- Run the code
- Test normal input
- Test unusual input
- Compare with requirements
- Ask tutor/peer
- Check documentation

Explain

Can I explain the final code without reading the AI response? What do I still need to understand?